

Statistical Models for Discrete Outcomes

Practical 1 Solutions

1. There are various ways to display summary statistics in Stata. The following is one such way.

```
. bysort treat : sum prement mentact
```

```
-> treat = Placebo Group
```

Variable	Obs	Mean	Std. dev.	Min	Max
prement	24	6	3.787852	0	14
mentact	24	5.541667	3.348967	0	11

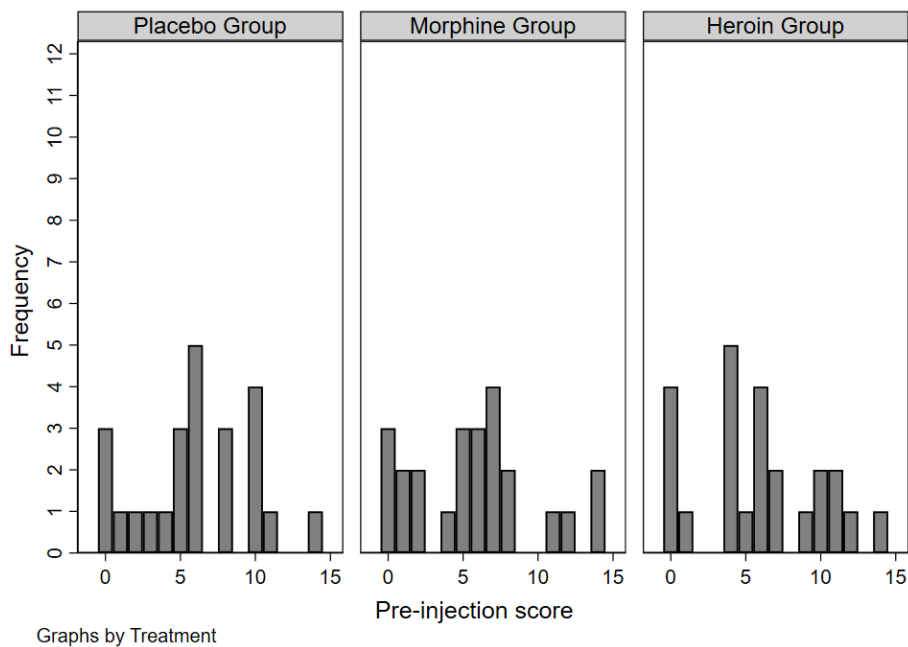
```
-> treat = Morphine Group
```

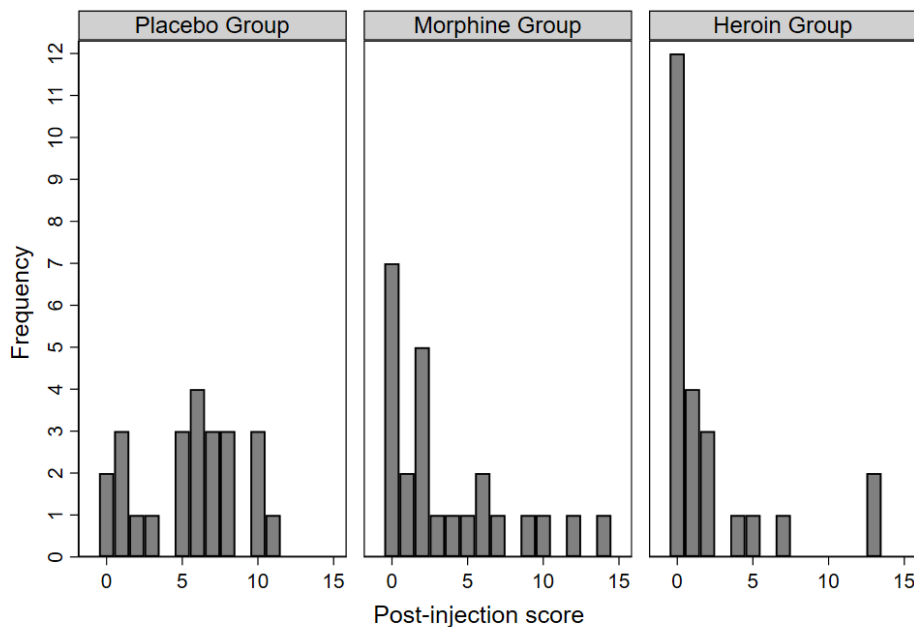
Variable	Obs	Mean	Std. dev.	Min	Max
prement	24	5.75	4.152003	0	14
mentact	24	3.666667	4.124863	0	14

```
-> treat = Heroin Group
```

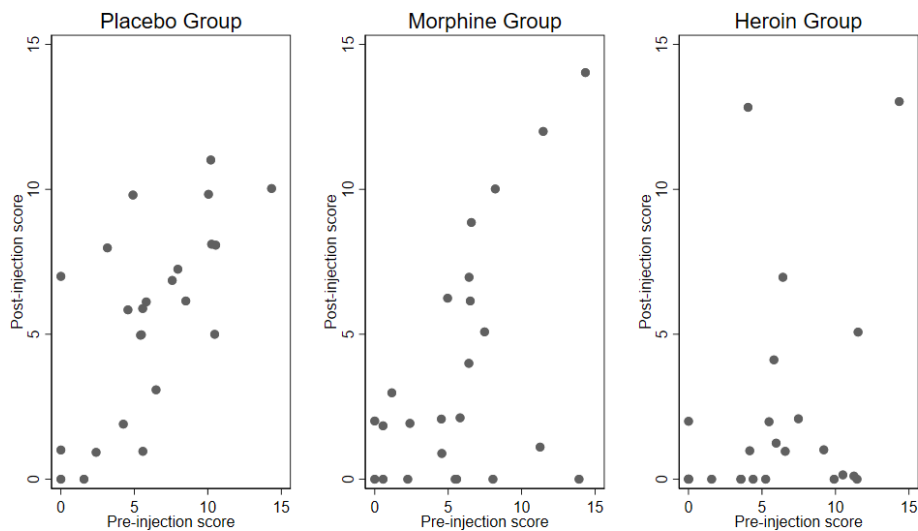
Variable	Obs	Mean	Std. dev.	Min	Max
prement	24	5.875	4.078709	0	14
mentact	24	2.166667	3.795497	0	13

The code to produce the plots below is given in the Do file.





Graphs by Treatment



Discussion: What conclusions can be drawn about the appropriateness of linear regression models for post-injection scores on the basis of these preliminary explorations?

The tabulations and plots indicate that the distributions of scores are similar pre-injection (as we would expect since this is a randomised study), but markedly different post-injection.

The histograms of pre- and post-injection scores suggest that these variables are not normally distributed. Note, however, that this does not necessarily invalidate inferences that we want

to draw from the regression models that we are going to fit, since in a regression model it is only residuals that need to be normally distributed for valid inferences.

The scatter plot shows that pre- and post-injection scores are positively associated. An assumption that the relationship between the two is linear in each group does not seem unreasonable.

2. Your completed table should look similar to that below.

Table 1

#	Linear predictor
1	$\eta_i = \beta_0$
2	$\eta_i = \beta_0 + \beta_1 u_i + \beta_2 v_i$
3	$\eta_i = \beta_0 + \beta_3 x_i$
4	$\eta_i = \beta_0 + \beta_1 u_i + \beta_2 v_i + \beta_3 x_i$
5	$\eta_i = \beta_0 + \beta_1 u_i + \beta_2 v_i + \beta_3 x_i + \beta_4 u_i x_i + \beta_5 v_i x_i$

Here x_i and Y_i are respectively the pre- and post-injection scores for individual i . The additional predictor variables are as follows.

$$u_i = \begin{cases} 0 & \text{if } i\text{th individual in Placebo or Heroin group} \\ 1 & \text{if } i\text{th individual in Morphine group} \end{cases}$$

$$\text{and } v_i = \begin{cases} 0 & \text{if } i\text{th individual in Placebo or Morphine group} \\ 1 & \text{if } i\text{th individual in Heroin group} \end{cases}$$

In each model the link function is the identity, $\eta_i = \mu_i$ and it is assumed that $Y_i \sim N(\mu_i, \sigma^2)$ and are independent.

3. Selected output from regression models

```
. ** Model 1
. regress mentact
```

Source	SS	df	MS	Number of obs	=	72
Model	0	0	.	F(0, 71)	=	0.00
Residual	1117.875	71	15.7447183	Prob > F	=	.
Total	1117.875	71	15.7447183	R-squared	=	0.0000
				Adj R-squared	=	0.0000
				Root MSE	=	3.968

```
. ** Model 2
. regress mentact i.treat
```

Source	SS	df	MS	Number of obs	=	72
Model	137.25	2	68.625	F(2, 69)	=	4.83
Residual	980.625	69	14.2119565	Prob > F	=	0.0109
Total	1117.875	71	15.7447183	R-squared	=	0.1228
				Adj R-squared	=	0.0974
				Root MSE	=	3.7699

```
. ** Model 3
. regress mentact prement
```

Source	SS	df	MS	Number of obs	=	72
Model	233.54726	1	233.54726	F(1, 70)	=	18.49
Residual	884.32774	70	12.6332534	Prob > F	=	0.0001
Total	1117.875	71	15.7447183	R-squared	=	0.2089
				Adj R-squared	=	0.1976
				Root MSE	=	3.5543

```
. ** Model 4
. regress mentact i.treat prement
```

Source	SS	df	MS	Number of obs	=	72
Model	365.819593	3	121.939864	F(3, 68)	=	11.03
Residual	752.055407	68	11.0596383	Prob > F	=	0.0000
Total	1117.875	71	15.7447183	R-squared	=	0.3272
				Adj R-squared	=	0.2976
				Root MSE	=	3.3256

```
. ** Model 5
. regress mentact i.treat#c.premment
```

Source	SS	df	MS	Number of obs	=	72
Model	384.747393	5	76.9494786	F(5, 66)	=	6.93
Residual	733.127607	66	11.107994	Prob > F	=	0.0000
Total	1117.875	71	15.7447183	R-squared	=	0.3442
				Adj R-squared	=	0.2945
				Root MSE	=	3.3329

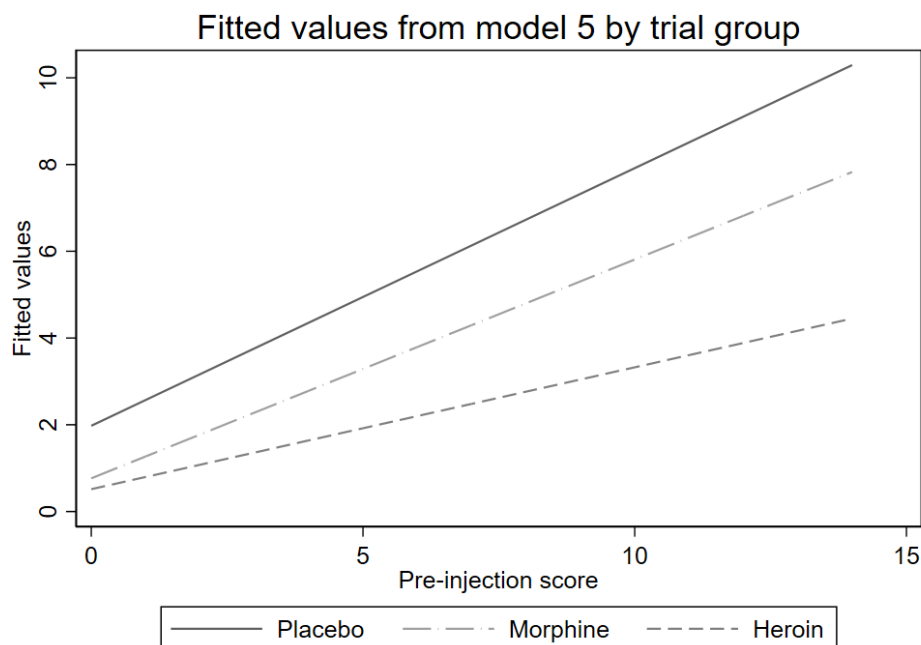
mentact	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
treat						
Morphine Group	-1.211742	1.750342	-0.69	0.491	-4.706413	2.282929
Heroin Group	-1.461968	1.771855	-0.83	0.412	-4.999591	2.075655
prement	.5939394	.1834682	3.24	0.002	.2276333	.9602455
treat#c.premment						
Morphine Group	-.0895258	.2483459	-0.36	0.720	-.5853644	.4063129
Heroin Group	-.3129855	.2503829	-1.25	0.216	-.8128911	.1869202
_cons	1.97803	1.294069	1.53	0.131	-.6056616	4.561722

The table below shows the number of parameters, the residual sums of squares, and the degrees of freedoms for each of the five models (in red above).

Table 2

#	Linear predictor	no. pars	RSS	rdf
1	$\eta_i = \beta_0$	1	1117.9	71
2	$\eta_i = \beta_0 + \beta_1 u_i + \beta_2 v_i$	3	980.6	69
3	$\eta_i = \beta_0 + \beta_3 x_i$	2	884.3	70
4	$\eta_i = \beta_0 + \beta_1 u_i + \beta_2 v_i + \beta_3 x_i$	4	752.1	68
5	$\eta_i = \beta_0 + \beta_1 u_i + \beta_2 v_i + \beta_3 x_i + \beta_4 u_i x_i + \beta_5 v_i x_i$	6	733.1	66

4. The code to produce the following plot is given in the Do file.



The fitted values from model 5 lie along three straight lines, one for each trial arm, with the slopes and intercepts being derived from the Stata output on the previous page as follows.

Table 3

Trial arm	Fitted intercept	Fitted slope
Placebo	1.97803	0.5939394
Morphine	$1.97803 + -1.211742 = 0.7662884$	$0.5939394 + -0.0895258 = 0.5044136$
Heroin	$1.97803 + -1.461968 = 0.5160623$	$0.5939394 + -0.3129855 = 0.2809539$

5. (a) To compare the fits of models 3 and 4 compute the relevant F-statistic from Table 2.

$$F = \frac{(884.3 - 752.1)/(70 - 68)}{752.1/68} = 5.98$$

Discussion: What do you conclude from this test?

Comparison with the $F_{2,68}$ distribution gives $p = 0.004$ ($0.001 < p < 0.005$ if using Neave's tables). There is therefore strong evidence of a difference in the post-injection scores between the groups after adjusting for pre-injection scores.

This test can also be carried out in Stata using the following command (after re-fitting model 4.).

```
. test 2.treat 3.treat

( 1)  2.treat = 0
( 2)  3.treat = 0

      F(  2,      68) =      5.98
      Prob > F =      0.0041
```

- (b) To compare the fits of models 4 and 5 compute the relevant F-statistic from Table 2.

$$F = \frac{(752.1 - 733.1)/(68 - 66)}{733.1/66} = 0.85$$

Discussion: What do you conclude from this test?

Comparison with the $F_{2,66}$ distribution gives $p = 0.43$ ($p > 0.1$ with Neave's tables). There is therefore no evidence of an interaction between group and pre-injection score.

This test can also be carried out in Stata (see Do file).

Discussion: Why can a F-test not be used to compare the fits of models 2 and 3?

Models 2 and 3 are not nested, so cannot be compared using an analogous approach to that above.

6.

```
. regress mentact i.treat prement
```

Source	SS	df	MS	Number of obs	=	72
Model	365.819593	3	121.939864	F(3, 68)	=	11.03
Residual	752.055407	68	11.0596383	Prob > F	=	0.0000
				R-squared	=	0.3272
				Adj R-squared	=	0.2976
Total	1117.875	71	15.7447183	Root MSE	=	3.3256

mentact	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
treat						
Morphine G..	-1.76151	.9603436	-1.83	0.071	-3.677845	.1548261
Heroin Group	-3.318255	.9601002	-3.46	0.001	-5.234105	-1.402405
prement	.4539615	.0998574	4.55	0.000	.2546991	.6532238
_cons	2.817898	.9054238	3.11	0.003	1.011153	4.624643

```
. glm mentact i.treat prement, fam(normal) link(id)
```

Iteration 0: Log likelihood = -186.62475

Generalized linear models	Number of obs	=	72
Optimization : ML	Residual df	=	68
	Scale parameter	=	11.05964
Deviance = 752.0554069	(1/df) Deviance	=	11.05964
Pearson = 752.0554069	(1/df) Pearson	=	11.05964

Variance function: V(u) = 1
Link function : g(u) = u

[Gaussian]
[Identity]

Log likelihood = -186.6247541	AIC	=	5.295132
	BIC	=	461.2421

		OIM				
mentact	Coefficient	std. err.	z	P> z	[95% conf. interval]	
treat						
Morphine Group	-1.76151	.9603436	-1.83	0.067	-3.643748	.1207291
Heroin Group	-3.318255	.9601002	-3.46	0.001	-5.200017	-1.436493
prement	.4539615	.0998574	4.55	0.000	.2582446	.6496783
_cons	2.817898	.9054238	3.11	0.002	1.0433	4.592496

The parameter estimates and standard errors are identical. However, the confidence intervals and p-values are slightly different. This is because the glm command approximates the exact t-distribution with the (asymptotically valid) z distribution. This is reported in the output table (heading of third column of results).

7. Key parameter estimates (and SEs) from models 2 & 4 are as follows.

Table 4

	Mean	Model 2		Model 4	
		Mean difference from placebo	SE	Adj. difference from placebo	SE
Placebo	5.542	0	–	0	–
Morphine	3.667	-1.875	1.088	-1.762	0.960
Heroin	2.167	-3.375	1.088	-3.318	0.960

8. **Discussion: Write a paragraph (or three!) to summarise your conclusions from these analyses.**

The F-test (from `reg mentact i.treat`) gives $p = 0.0109$, indicating moderately strong evidence that the groups differ in respect of their post-injection scores. Looking at the p-values corresponding to the individual coefficients, we see that the estimated difference between morphine and placebo is not formally statistically significant at the 5% level, but is close ($p = 0.089$), whilst the estimated difference between heroin and placebo is formally statistically significant ($p = 0.003$).

Adjusting for “prement” changes the estimated group differences somewhat, but not dramatically. This is as we would expect. Since the trial is randomized there is no confounding. There are only small chance differences between the group means at baseline. Adjusting for these changes the estimated treatment effects slightly, but not materially. The primary purpose of baseline adjustment in randomised trials is to improve the efficiency of treatment effect estimates, this being reflected in (here only slightly) reduced standard errors.

The F-test for the interaction parameters in the last model gives $p = 0.43$, indicating no evidence against the null hypothesis that the effects of the drugs are the same irrespective of pre-injection mental activity. So, we cannot say that there is evidence that the effects of the drugs are different for subjects with different levels of pre-injection mental activity.

9. (a) Using lincom

After fitting model 4 the lincom command can be usefully used to obtain a 95% confidence interval for the difference in the post-injection scores on heroin and morphine after adjusting for pre-injection scores.

```
. lincom 3.treat - 2.treat
```

```
( 1) - 2.treat + 3.treat = 0
```

	mentact	Coefficient	Std. err.	t	P> t	[95% conf. interval]
(1)		-1.556745	.9601002	-1.62	0.110	-3.472595 .3591049

(b) Or re-parameterise to change the reference category to the second level of the treatment variable, which corresponds to the morphine group.

```
. regress mentact b2.treat prement
```

	mentact	Coefficient	Std. err.	t	P> t	[95% conf. interval]
treat						
Placebo Group		1.76151	.9603436	1.83	0.071	-.1548261 3.677845
Heroin Group		-1.556745	.9601002	-1.62	0.110	-3.472595 .3591049
prement		.4539615	.0998574	4.55	0.000	.2546991 .6532238
_cons		1.056388	.8891012	1.19	0.239	-.7177855 2.830562

The coefficient for the heroin group now represents the difference between the morphine and heroin groups.